Heating of planetesimals from ^{60}Fe & ^{26}Al

Hunting the source of short-lived radioisotopes and simulating desiccation in planetesimals

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Contents & questions

What are SLRs?

Why are they important?

How do they end up in protoplanetary disks?

How do disks survive this enrichment?

How do they affect planet formation?

[1] Fu & Elkins-Tanton 2014

What are SLRs?

Short-lived radioisotopes

- Is what it says on the tin!
- Half-lives on the order of 1 Myr.
- ^{26}Al and ^{60}Fe primarily discussed.
- Decay of SLRs constituted majority of heating in early solar system [1].
- Homogenous throughout solar system.
- Wolf-Rayet (WR) winds and supernovae are sources of SLRs, though there are some issues with both.

[1] Fu & Elkins-Tanton 2014

Planetesimal desiccation

- Decay heating in planetesimals causes vaporisation and outgassing [1].
- This leads to desiccation & eventual formation of water-poor planets.
- SLR heating also keeps planetesimal internals molten, helps with stratification.

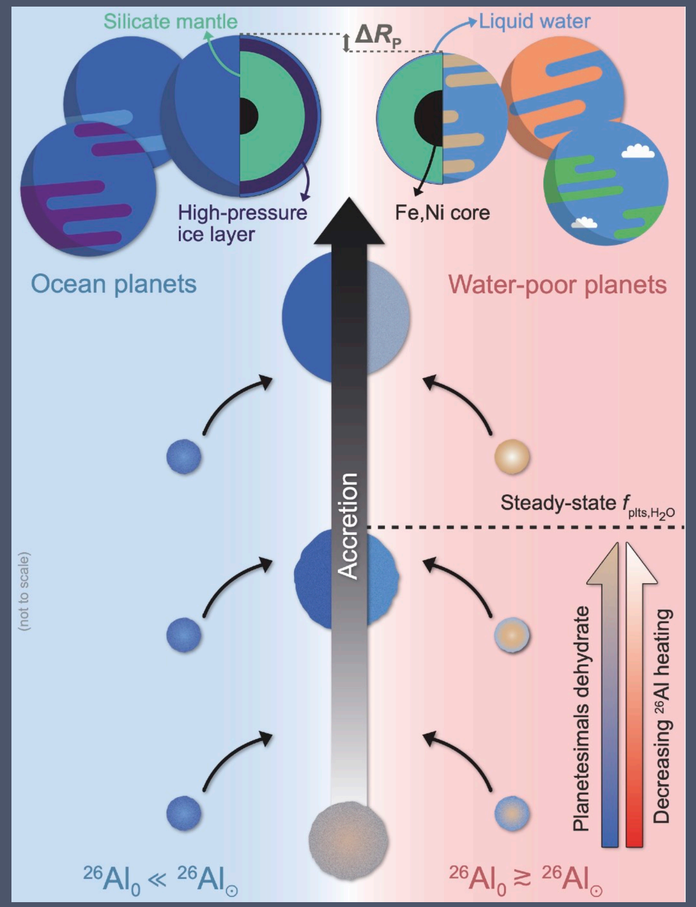


Fig. Lichtenberg+ 2019, [1] Lichtenberg+ 2019

SLR enrichment of disks

Motivations

How common are highly enriched ^{26}Al and ^{60}Fe disks?

How dependent is enrichment on star forming region density?

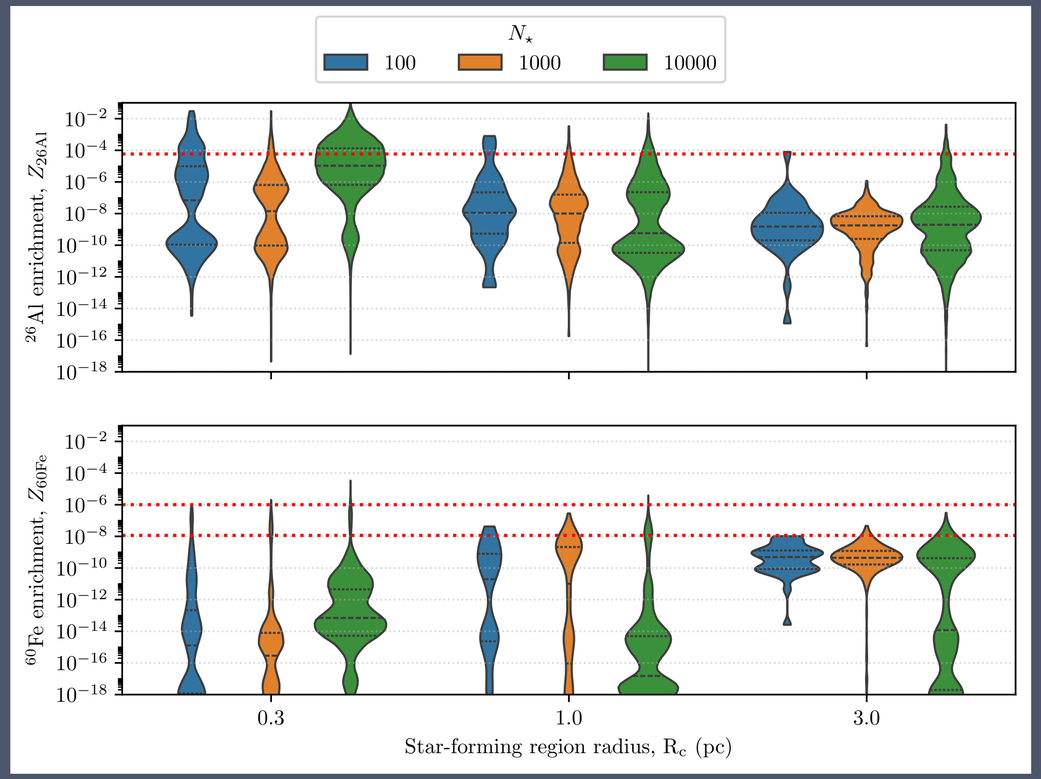
Are disks enriched through SNe or early-type stellar winds?

Is there another pre-formation mechanism?

Enriching disks

Eatson, Parker & Lichtenberg 2024

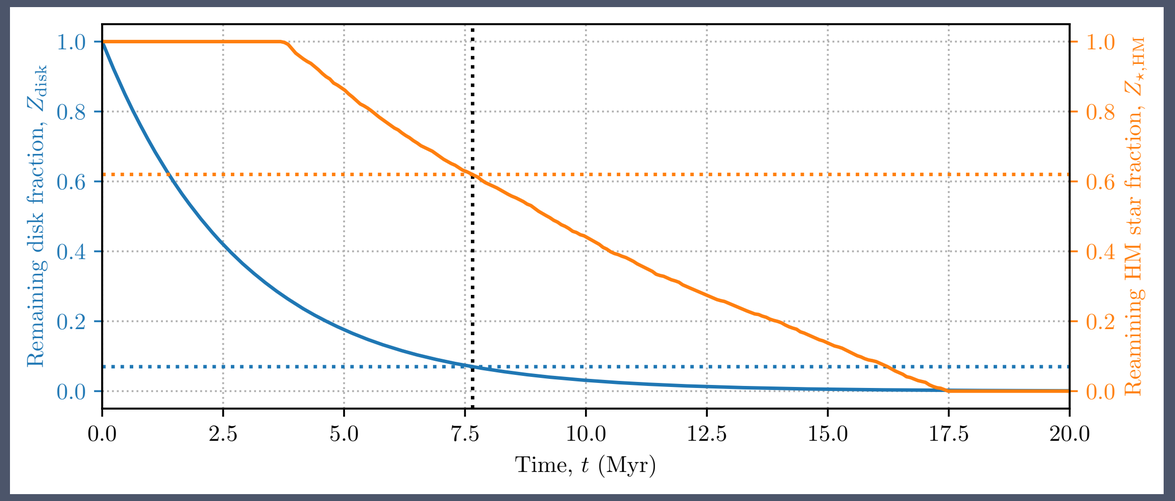
- AMUSE [1] library used to combined N-body and stellar evolution codes.
- Enrichment dependent on median local density of cluster.
- ^{26}Al enrichment above solar system levels somewhat common.
- ^{60}Fe enrichment is less pronounced.



[1] Portegies Zwart & McMillan (2018)

The hitch with winds and SNe

- ^{60}Fe not produced substantially by WR winds [1].
- Insufficient population of disks when supernovae occur.
  - "Islands of explodability".
- Photoevaporation and shocks [2].
- Other methods of production?



X-axis: time
Y-axis left/blue: Remaining disks
Y-axis right/orange: Remaining high-mass stars

[1] Limongi & Chieffi 2018 [2] Patel+ 2023

Interlopers?

- AGB star observed to pass through cluster [1].
- Produce copious amounts of ^{26}Al *and* ^{60}Fe.
- Gentler winds, less UV flux, less likely to damage disk.

[1] Parker & Schoettler, 2023

Interlopers...

How sensitive would interloper enrichment be:

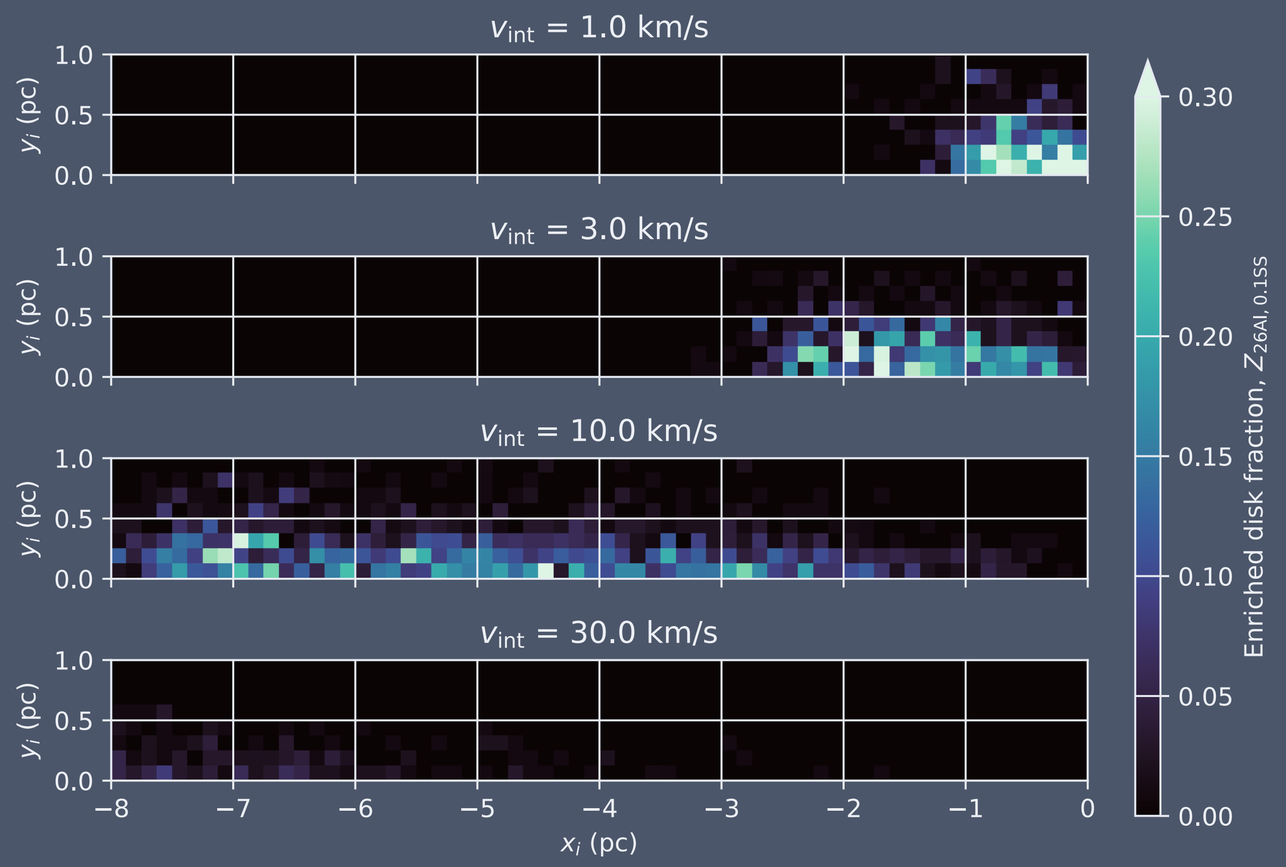
To AGB evolution?

To interloper velocity?

Interlopers!

Eatson & Parker
In prep.
QR code for poster!

- ~30% of disks well enriched in best case.
- "Near misses" also acceptable.
- Faster interlopers less sensitive to initial distance.
- Enrichment even at 30 km/s.
- Good ^{60}Fe enrichment too!



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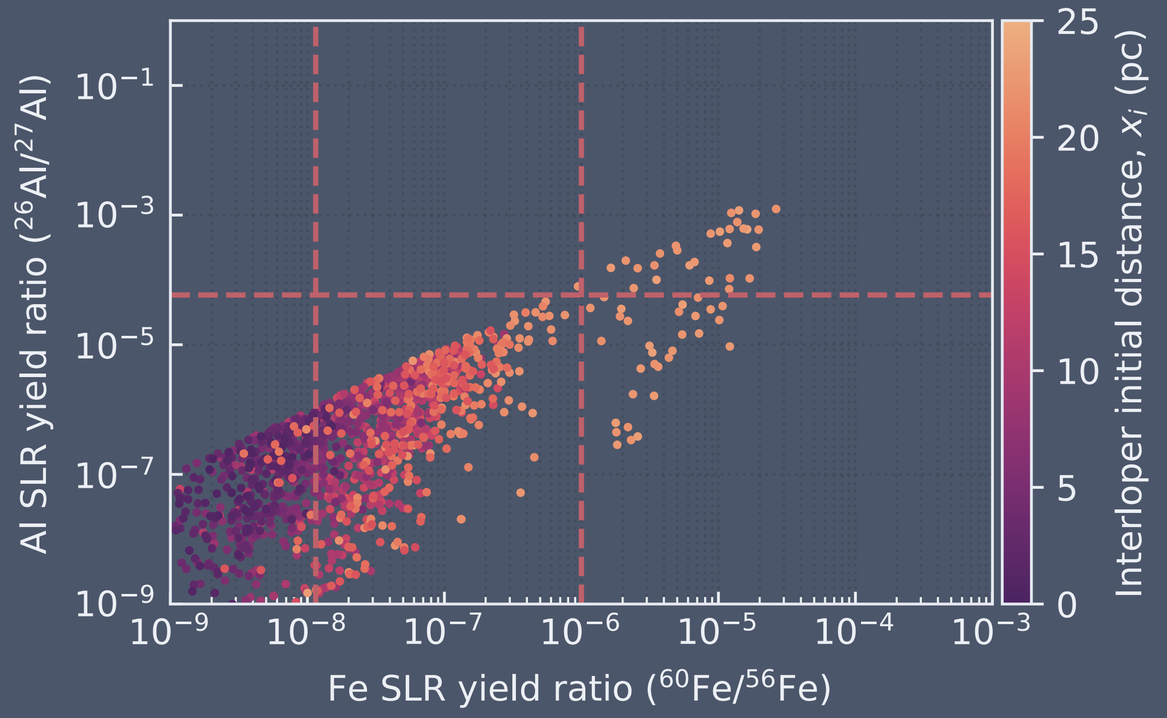


Planetesimal heating by SLRs

More motivations

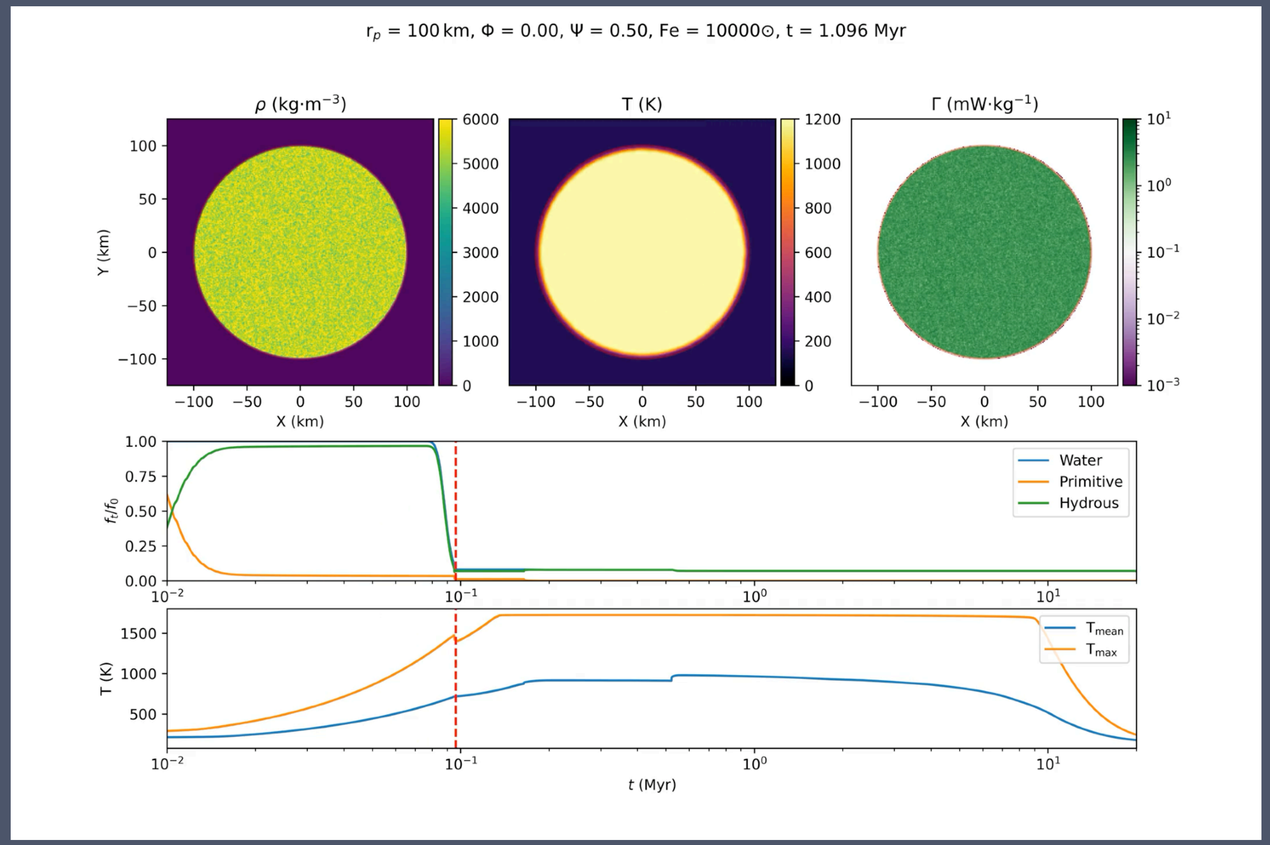
How much does the SLR ^{60}Fe influence heating and desiccation?

How much ^{60}Fe needed to get desiccation?

Melting rocks

Eatson, Parker, Lichtenberg & Gerya 2024

- Series of simulations using I2ELVIS 2D geodynamic numerical code **[1]**.
- Parameter space explored varying ^{26}Al & ^{60}Fe enrichment and Fe content.
- Measuring water retention fraction.

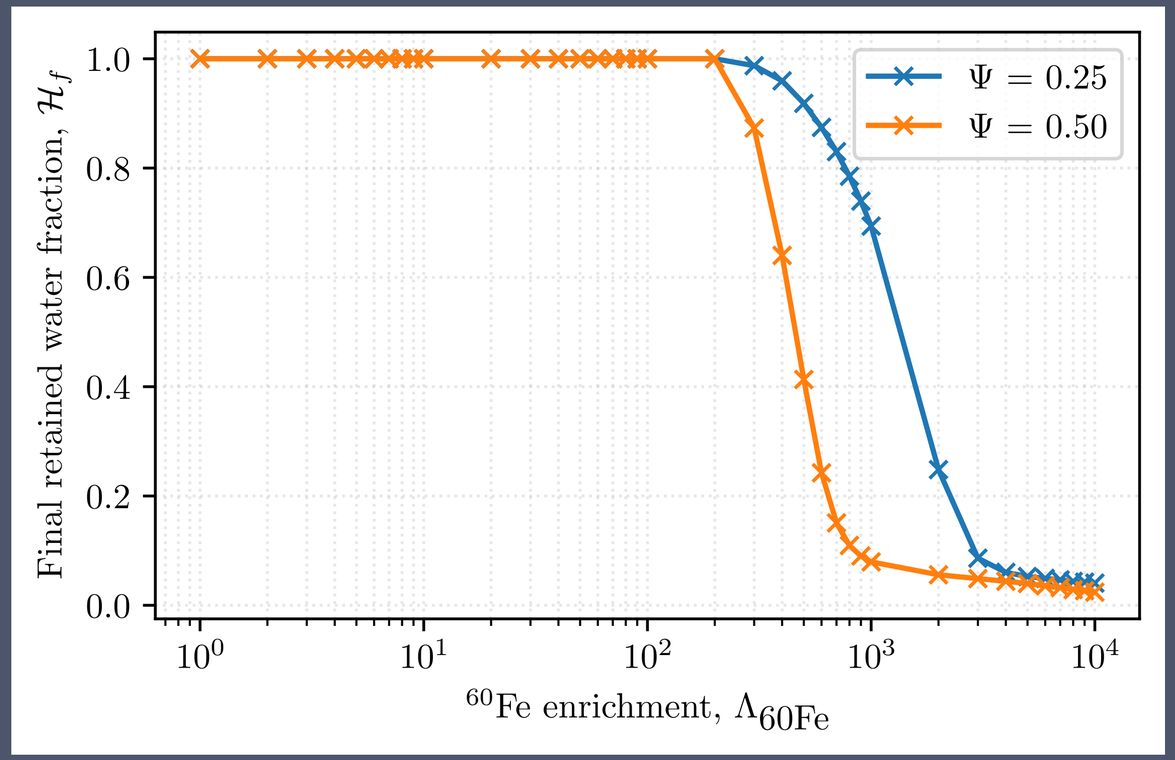


[1] Gerya (2019), `github.com/FormingWorlds/i2elvis_planet`

Drying out rocks

Eatson, Parker, Lichtenberg & Gerya 2024

- Desiccation occurs extremely rapidly after simulation onset.
- ^{60}Fe ~200x higher than solar system levels for any desiccation!
- 1/10th solar needed for ^{26}Al **[1]**.
- ^{26}Al enrichment levels reachable for AGBs.



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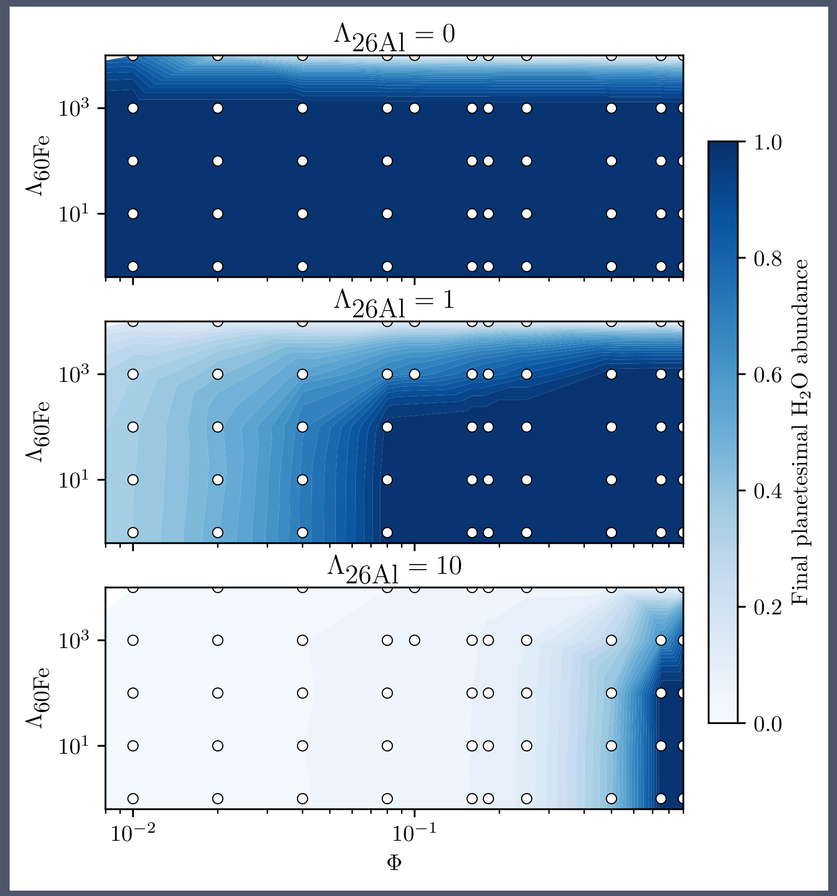

[1] Lichtenberg+ 2019

Probably gone long

Wrap it up!

Conclusions

- Decay heat from SLRs in disks causes desiccation of planetesimals.
- Decay heating mainly through ^{26}Al, sims show ^{60}Fe needs high enrichment.
- *N*-body sims show that ^{60}Fe enrichment levels leading to desiccation less.
- Enrichment through interlopers provides an alternate, gentler route to disk enrichment.

Questions?

⇦ *Slide deck and poster here!*

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